QuantERA Call 2025 Proposal Initial Draft

1. Excellence (max. 6 pages)

1.1 Targeted breakthrough, baseline of knowledge and skills

Describe the targeted breakthroughs of the project.
Describe how the science and technology contribute to the establishment of a solid baseline of knowledge and skills for the specific theme addressed.
Describe the specific objectives of the project, which should be clear, measurable, realistic and achievable within the duration of the project.

Targeted Breakthroughs

The central ambition of this project is to overcome the three fundamental barriers currently preventing Quantum Machine Learning (QML) from surpassing classical capabilities: **Scalability** (limited qubit connectivity and count), **Trainability** (the optimization trilemma of barren plateaus, measurement overhead, and classical simulability), **Temporal Expressibility** (inability to model long-range correlations), and **Reliability** (inherent NISQ fragility and lack of robustness).

Current QML research is often limited to small-scale "toy" problems on single processors, heavily reliant on heuristic ansatzes that become untrainable or unreliable at scale. Our targeted breakthrough is to shift the paradigm from "heuristic QML" to **"Physics-Aware & Reliable QML,"** establishing a rigorous theoretical and algorithmic foundation that exploits quantum resources efficiently and robustly.

We target three specific foundational breakthroughs:

1. **Foundational Scalability via Multi-Chip Ensembles:** We aim to break the monolithic chip limit by developing **Multi-Chip Ensembles** using distributed quantum computing and classical ensemble learning techniques. Rather than waiting for large-scale hardware, we will utilize distributed QML architectures to demonstrate that entanglement resources can be effectively distributed across separated quantum processing units (QPUs).
   Beyond standard parallelization, we target **Multi-Modal Data Fusion**: using separate quantum processors to encode heterogeneous data types (e.g., Chip A for sMRI, Chip B fMRI). We will demonstrate that entangled ensemble strategies can fuse these distinct quantum feature spaces to achieve a "Collective Quantum Advantage" unavailable to single-chip models.
2. **Solving the Trainability-Efficiency-Advantage Trade-off via Physics-Informed Optimization:** We target the fundamental trade-offs limiting QML optimization: **Barren Plateaus** (untrainability in deep circuits), **Measurement Overhead** (prohibitive sampling costs for both gradient and gradient-free methods), and the **Simulability Trap** (where trainable models lose quantum advantage). We will implement a hybrid **Quantum Forward-Forward (QFF)** algorithm driven by **Hybrid Quantum Genetic Algorithm (HQGA)**.
   - **Addressing Trainability & Advantage:** QFF utilizes **local goodness measurements** to bypass Barren Plateaus without restricting the circuit to classically simulable log-depth structures, thereby preserving **Quantum Advantage.**

- **Addressing Efficiency:** By decoupling layers, QFF avoids the costly backpropagation of global error. The use of **superposition and entanglement** in the HQGA significantly speed-up the convergence of conventional classical evolutionary algorithms, balancing the exploration-exploitation trade-off to minimize measurement overhead.

3. **Next-Generation Temporal & Structural Learning:** We target the limitations of current QML in processing time-series and graph data. We will develop **Quantum State Space Models (Q-SSM)** that can effectively learn complex, high-dimensional long-range dependencies in sequential data by leveraging quantum mechanics' inherent properties. The quantum circuits exploit entanglement to create **non-local correlations across feature dimensions in a single operation**—capturing dependencies that classical models require deep stacking or attention mechanisms to achieve—while the LSTM-style gating propagates this information across arbitrarily long sequences. Unlike classical transformers that require quadratic computational complexity $O(L^2)$ where L is sequence length, our Q-SSM model has **linear complexity** $O(L)$ and access to a 2^n-dimensional Hilbert space representation. Additionally, the quantum circuits achieve this expressivity with only $O(n \times l)$ parameters versus $O(N^2)$ for equivalent classical layers (n: number of qubits; l: number of quantum circuit layers; N: hidden dimension size), making the models particularly **suitable for data-limited domains** like biomedical signal processing where capturing **subtle long-range temporal patterns** (e.g., brain state dynamics across seconds of EEG) is critical but **overfitting** is a major concern.
4. **Robustness & Reliability via Fuzzy Logic and Benchmarking Protocols:** We aim a paradigm shift from *suppressing* quantum noise to *exploiting* it as a physical generative resource. By utilizing **Fuzzy Quantum Logic** to bridge the mathematical gap between discrete qubits and continuous neural networks, we develop **Physics-Informed Fuzzy Quantum Diffusion Model** that achieves the first "Hardware-Native" AI architecture that learns the exact, drifting physical fingerprint of a specific quantum device. This enables NISQ computers to perform high-fidelity generative tasks and bespoke error mitigation without waiting for fully fault-tolerant hardware, effectively turning the intrinsic instability of quantum systems into a unique computational feature rather than a liability. To curb uncertainty from noisy models even further, we will design **benchmarking and certification protocols**, leveraging the **QUARK** (QUantum computing Application benchmaRK) framework, to ensure they meet **industrial-grade robustness** standards against adversarial perturbations and NISQ noise. QUARK enables this by providing a standardized, vendor-neutral platform to rigorously quantify critical performance metrics—such as **noise resilience**, **trainability**, and **expressibility**—under realistic hardware conditions. This systematic validation allows us to definitively prove that our Fuzzy Quantum Diffusion architecture successfully delivers a genuine quantum advantage over classical baselines.

Baseline of Knowledge and Skills

This project advances the state-of-the-art by establishing a new baseline for **reliable, scalable, and trainable QML**.

- **From Monolithic to Distributed Scalability:** Currently, QML is restricted to small, unimodal tasks. We establish a baseline for **Distributed Multi-Modal Quantum Ensembles**, providing the protocols to fuse heterogeneous data (e.g., sMRI+fMRI) across networked QPUs via classical aggregation strategies.
- **From Gradient-Dependent to Gradient-Free/Hybrid Learning:** The community currently struggles with the convergence of Variational Quantum Algorithms (VQAs). By establishing the efficacy of **QFF combined with Hybrid Quantum Genetic**

Algorithms, we provide a new baseline for training deep quantum networks, proving that learnability can be maintained where classical gradients fail.
- **From Static Snapshots to Long-Sequence Spatio-Temporal Learning:** Current QML approaches struggle to process dynamic, sequential data efficiently. By establishing **Q-SSM** as standard tools, we provide a new baseline for **learning spatio-temporal data of long sequences**, enabling the analysis of high-dimensional biological signals with linear computational complexity.
- **From Fragile to Certified Robustness:** By combining **Fuzzy Logic** strategies with the **benchmarking and certification protocols**, we establish a methodology for **Reliable QML**. This contributes to the baseline skill set needed to certify quantum algorithms for safety-critical applications.

The consortium brings together a unique, synergistic baseline of skills:

- **QML Architectures & Scalability (SNU - Cha):** Deep expertise in **Multi-Chip Ensembles**, **Spatio-temporal QML algorithms, Quantum State Space Models (Q-SSM),** and computational neuroscience.
- **Optimization & Logic (Univ. Naples - Acampora):** World-class expertise in Computational Intelligence, Evolutionary Algorithms, and Fuzzy Logic.
- **Reliability & Certification (Fraunhofer IKS - Lorenz):** Industrial-grade, **application-oriented QML** expertise focused on safety, reliability and data efficiency aspects.
- **Domain Validation (Yonsei Univ - Yoo):** Experimental High Energy Physics (HEP) expertise, providing the massive, high-dimensional datasets required to rigorously stress-test the quantum advantage.

Specific Objectives

The project is structured around four clear, measurable, realistic, and achievable objectives:

- **Objective 1: Demonstrate Scalable Distributed QML (The Scalability Objective)**
  - **Goal:** Develop **Multi-Chip Ensembles** using input partition and ensemble learning protocols.
  - **Measure:** Successfully train a QML model (e.g., Quantum Transformer) fusing sMRI and fMRI features distributed across at least **2 simulated QPUs** (and physical hardware if available).
  - **Target:** Demonstrate **Scalability** by processing a multi-modal dataset that exceeds the qubit capacity of a single available chip (simulating a **2x capacity increase** via distributed processing) while achieving **>90% classification accuracy** on neuroimaging benchmarks.
  - **Timeline:** Protocols defined by Month 12; Full validation by Month 30.
- **Objective 2: Solve the Trainability-Efficiency-Advantage Trade-off (The Trainability Objective)**
  - **Goal:** Develop the **Quantum Forward-Forward (QFF)** algorithm hybridized with **Hybrid Quantum Genetic Algorithms**.
  - **Measure:** Compare convergence rates, measurement overhead, and classical simulability against standard Backpropagation and pure Evolutionary Algorithms.
  - **Target:** Achieve **convergence** in deep circuits (>10 layers) where Backpropagation fails (Barren Plateaus), with a **20% reduction** in measurement overhead compared to parameter-shift rules, while maintaining non-classicality.
  - **Timeline:** Algorithm design by Month 12; Benchmarking by Month 24.
- **Objective 3: Advance Next-Gen Temporal & Structural Learning (The Temporal Expressibility Objective)**

    - **Goal:** Develop **Q-SSM** for high dimensional spatio-temporal data.
    - **Measure:** Compare **Q-SSM** prediction accuracy on long sequence EEG/fMRI data against classical transformers/SSMs and standard Quantum RNNs.
    - **Target:** Demonstrate **superior long-range dependency capture** (measured by memory capacity metrics) and linear scaling with sequence length.
    - **Timeline:** Theoretical framework by Month 18; Implementation by Month 30.
- **Objective 4: Tackling Uncertainty and Unreliability of Noisy Quantum Devices (The Reliability Objective)**
    - **Goal:** Create a **Fuzzy Quantum Diffusion Model** enhanced by Fuzzy Logic controllers and reduce the model's uncertainty by **certifying its robustness** with **QUARK** framework.
    - **Measure:** Generate high-fidelity synthetic data (measured by Fréchet Inception Distance - FID) under simulated NISQ noise and assess robustness using reliability metrics (e.g., Lipschitz continuity).
    - **Target:** Achieve superior generative performance and certified robustness compared to classical transformers/diffusion models and standard quantum generative models (e.g., Quantum GAN) under noise levels characteristic of current hardware.
    - **Timeline:** Theoretical framework by Month 18; Implementation by Month 36.
- **Objective 5: Validate Foundational Advantage via Domain Science (The Validation Objective)**
    - **Goal:** Use high-complexity domain data to certify the foundational breakthroughs (O1-O4).
    - **Measure:**
        - **High Energy Physics (HEP):** Apply **Multi-Chip Ensembles** to CMS calorimeter-image classification (Quantum Vision Transformer), feature-based background estimation (Q-ABCDisCo), and waveform denoising (Temporal Convolutional Network-VQC) to assess scalability and quantum advantage across multiple HEP modalities.
        - **Neuroscience:** Apply **Quantum Transformers** and **Q-SSM** to EEG/fMRI data, and **QFF-Evolutionary VQE/QAOA** for brain connectivity analysis.
        - **Cybersecurity:** Apply **Robust Fuzzy-Quantum Diffusion Models** to intrusion detection logs and certify robustness against cyber- and privacy attacks via comprehensive **benchmarking protocols**.
    - **Target:** Demonstrate quantum advantage in either accuracy, expressibility, or training efficiency compared to classical SOTA on these specific datasets.
    - **Timeline:** Data preparation by Month 6; Final Domain Validation by Month 36.

1.2 Novelty, level of ambition and foundational character

Describe the advance your proposal would provide beyond the state-of-the-art, and to what extent the proposed work is ambitious, novel and of a foundational nature. Your answer could refer to the ground-breaking nature of the objectives, concepts involved, issues and problems to be addressed, and approaches and methods to be used.

Advance Beyond the State-of-the-Art

Current Quantum Machine Learning (QML) is hitting four "hard walls": the **Hardware Wall** (single chips are too small), the **Optimization Wall** (Barren Plateaus make deep circuits untrainable), **Temporal Expressibility Wall** (long range temporal modeling is inefficient), and the **Reliability Wall** (models collapse under noise and lack industrial certification). This project proposes a foundational shift to dismantle these walls, moving beyond the current state-of-the-art (SOTA) as follows:

1. **From Monolithic to Distributed Multi-Modal QML Scalability:**
   - **State-of-the-Art:** Current QML is constrained to single QPUs and simple datasets of single modality. Scaling requires larger fault-tolerant devices, which are years away. "Distributed" approaches often rely on expensive circuit cutting that incurs heavy sampling overhead.
   - **Our Novelty (Multi-Chip Ensembles):** We introduce a **Distributed Multi-Modal Ensemble** framework, which aggregates independent quantum models trained on heterogeneous data types (e.g., sMRI on Chip A, fMRI on Chip B). Instead of cutting circuits, our **Multi-Chip Ensembles** framework utilizes input partition and classical ensemble learning techniques to knit together multiple smaller QPUs. We specifically optimize the "inter-chip entanglement cost" to train high-expressibility models (such as Quantum Transformers) across fragmented hardware without requiring a single large fault-tolerant device.
   - **Foundational Advance:** We move beyond simple parallelization to **Multi-Modal Data Fusion** in Hilbert space. We establish the resource theory for **"Collective Quantum Advantage,"** demonstrating that entangled ensemble strategies can fuse distinct quantum feature spaces to solve problems larger than any single available chip.
2. **From Global Backpropagation to Local and Evolutionary Learning:**
   - **State-of-the-Art:** Optimization in QML faces a trilemma. Gradient-based methods suffer from **barren plateaus** and **high measurement costs**. Gradient-free methods suffer from **slow convergence**. Strategies that "fix" trainability (like restricting circuit depth) often render the model **classically simulable**, destroying quantum advantage.
   - **Our Novelty (Quantum Forward-Forward & Hybrid Quantum Genetic Algorithms):** We introduces a novel symbiotic framework that integrates the architectural decoupling of the **Quantum Forward-Forward** algorithm with the quantum-native evolutionary mechanics of the **Hybrid Quantum Genetic Algorithm (HQGA)**. Unlike state-of-the-art methods that are forced to choose between the prohibitive measurement overhead of global backpropagation or the "dequantization" required to avoid barren plateaus, our approach foundationally redefines QML training by decomposing deep circuits into **locally optimized layers** solved via **entangled crossover** and **quantum elitism**. This represents a paradigm shift that allows for the training of deep, highly entangled quantum models on NISQ hardware—architectures that are currently deemed untrainable—by mathematically circumventing the vanishing gradient problem while preserving the non-classical computational advantage that gradient-free methods typically sacrifice.
   - **Foundational Advance:** We demonstrate that this hybrid approach navigates the "Classical Simulability Trap." By using HQGA's global search, we can train deep, expressive circuits that remain **mathematically non-simulable** by classical computers, preserving the core quantum resource.
3. **From Static Snapshots to Next Generation Spatio-Temporal Learning:**
   - **State-of-the-Art:** Current QML algorithms are largely static. Existing Quantum RNNs struggle with high dimensional spatio-temporal data with long sequences. While classical transformers can effectively learn complex, long range dependencies, they incur a high computational cost that scales quadratically $O(L^2)$ with sequence length L.
   - **Our Novelty (Q-SSM):** We develop **Quantum State Space Models (Q-SSM)** architecture that integrates variational quantum circuits with recurrent gating mechanisms for sequential data, addressing a critical gap in existing QML—prior quantum sequence models either lacked temporal memory (collapsing sequences

via pooling) or attempted fully quantum recurrence that breaks gradient flow. Our hybrid design uniquely combines three-branch quantum superposition for expressive feature extraction with classical LSTM-style gating for selective memory, enabling **end-to-end differentiable training while preserving quantum expressivity**.

   - **Foundational Advance:** We set a new paradigm for spatio-temporal QML algorithms that exploits quantum entanglement for capturing **correlations across high-dimensional feature spaces** that are provably hard for classical models to represent efficiently. By demonstrating that quantum circuits can serve as drop-in replacements for classical feature extractors within proven recurrent architectures, we provide a practical template for quantum-classical hybrid design that bridges near-term quantum hardware capabilities with real-world sequence modeling tasks.

4. **From Noise-induced Uncertainty to Robustness:**
   - **State-of-the-Art:** In current NISQ devices, noise is treated as a binary error to be corrected via resource-heavy error correction codes. Generative QML models (e.g., Quantum GANs) typically fail on NISQ devices because hardware noise distorts probability distributions. Benchmarking and certifying QML methods is becoming increasingly important for deployment for real-world applications. First steps towards benchmarking have already been undertaken with the QUARK benchmarking framework. However, the industrial standards for certifying robustness against cyberattacks for QML are currently lacking.
   - **Our Novelty (Fuzzy-Quantum Diffusion & Robustness Certification):** We introduce **Fuzzy Quantum Logic** to handle hardware noise not as an "error" to be corrected, but as a "degree of uncertainty" within the learning process. Crucially, we move beyond simple performance metrics by developing **benchmarking and robustness protocols** to rigorously certify the reliability and robustness of these models against adversarial perturbations and noise, defining a new industrial standard for Trustworthy QML.
   - **Foundational Advance:** We establish the first **"QML Reliability Standard."** By defining formal metrics (e.g., Lipschitz continuity, noise stability) based on the QUARK framework, we transform QML from an experimental art into a **certified engineering discipline** capable of safety-critical deployment.

Level of Ambition and Foundational Character

This proposal is highly ambitious as it does not merely apply existing algorithms to new data; it rewrites the **foundational protocols** of how quantum models are built, trained, and verified.

- **Foundational Scalability:**
  We address the foundational question: "How much entanglement is required to link distinct quantum processors?" By developing **Multi-Chip Ensemble** protocols, we are establishing a resource theory for distributed quantum learning. This is ground-breaking because it decouples QML progress from the slow roadmap of monolithic hardware scaling, enabling "Virtual Large-Scale QML" on today's fragmented NISQ machines.
- **Ambition in Complexity (The "Big Science" Validation):**
  We are not testing on toy datasets (e.g., MNIST). We are ambitiously targeting High Energy Physics (LHC data) and Neuroscience (EEG/fMRI).
  - **HEP:** CMS detector data—including calorimeter images, high-level physics features, and waveform signals—contain complex, high-dimensional structures that classical algorithms often fail to disentangle. We aim to prove that our **Multi-Chip Ensembles** can capture non-local correlations in particle jets that classical deep learning misses.
  - **Neuroscience:** Brain signals are non-stationary and exhibit complex graph

structures. We aim to demonstrate that **Quantum State Space Models (Q-SSM)** can decode long-range temporal patterns with higher fidelity than classical SOTA. Furthermore, we will apply **QFF-HQGA VQE/QAOA** to solve combinatorial brain connectivity problems, identifying functional graph structures that classical heuristics fail to optimize.

- **Foundational Trust (The "QUARK Reliability" Standard):**
  We move beyond "accuracy" to "**certified robustness**." We are establishing a foundational methodology for benchmarking the safety and reliability of QML algorithms against adversarial attacks in **Cybersecurity** contexts. This establishes the first rigorous "Quality Assurance" framework for foundational QML resources, a critical step for the QPR topic and beyond.

Issues and Problems Addressed

Critical Issue	**Proposed Foundational Solution**
Connectivity & Scale Limits: Single chips lack the qubit count and connectivity for high-dimensional data.	**Multi-Modal Multi-Chip Ensembles:** Distributed protocols that aggregate parallel quantum agents to process heterogeneous data (sMRI/fMRI) without requiring a monolithic fault-tolerant device.
The Trainability Trilemma: Gradient-based methods suffer from Barren Plateaus and high measurement costs; gradient-free methods suffer from slow convergence.	**Hybrid Physics-Informed Optimization:** A synergy of **Quantum Forward-Forward (QFF)** (local updates to bypass Barren Plateaus) and **Hybrid Quantum Genetic Algorithms** (eliminates measurement overhead and classical simulability), balancing efficiency with trainability.
Inefficient Temporal Modeling: Classical transformers are computationally expensive, scaling quadratically to sequence length. Quantum RNNs struggle to learn high dimensional, long dependencies.	**Quantum State Space Models (Q-SSM): Three-branch quantum superposition** for rich feature extraction and LSTM-style gating for **selective memory management**, which enables efficiently and effectively learning long sequence data.
Lack of Trust & Robustness: NISQ models are fragile to noise and lack industrial safety certification.	**QUARK-Certified Fuzzy Robustness:** Integrating **Fuzzy Logic** to model noise as uncertainty, rigorously benchmarked against the **QUARK reliability standards** to certify robustness against adversarial attacks.

In summary, this project is foundational because it creates the **"Operating System" for Reliable and Scalable QML**: a suite of protocols (Distributed Ensembles, Evolutionary Optimization, QUARK Reliability) that allows quantum software to run effectively on imperfect hardware, validated by the most complex scientific data available.

1.3 Concept and methodology

Describe and explain the overall concept and research approach underpinning the project. Describe the main ideas, models or assumptions involved. Identify any interdisciplinary considerations and, where relevant, use of stakeholder knowledge.
Describe any national or international research and innovation activities which will be linked with the project, especially where the outputs from these will feed into the project.

Describe the methodology and explain its relevance to the objectives.
Describe the appropriateness of the methodology to narrow down multiple options and to address high scientific and technological risks.

Overall Concept: The "Physics-Aware" QML Stack

The overarching concept of this project is to transition Quantum Machine Learning (QML) from its current heuristic, hardware-agnostic state to a **"Physics-Aware" QML Stack**. We operate on the core assumption that Near-Term Intermediate Scale Quantum (NISQ) devices will remain fragmented (limited connectivity) and noisy for the foreseeable future. Therefore, instead of waiting for fault tolerant hardware, we fundamentally redesign the *software and algorithmic layer* to exploit quantum resources under these constraints.

Our research approach is built on a **Feedback Loop between Foundational Theory and Domain Science**, addressing five specific objectives:

1. **Scalability:** The **Multi-modal Multi-Chip Ensembles** framework addresses the scalability constraints of NISQ hardware by partitioning high-dimensional multi-modal data across independent quantum processors, fundamentally operating on the premise that restricting global entanglement mitigates barren plateaus and enhances noise resilience through the classical aggregation of independent error profiles. This approach **assigns distinct data modalities** (e.g., sMRI, fMRI) **to specific chips** equipped with **orthogonal circuit architectures** to maximize feature distinctiveness without requiring lossy dimension reduction. By employing a "**Selective Entanglement**" mechanism, the model strategically introduces quantum connections only between chips processing globally dependent features, effectively optimizing the trade-off between necessary model expressibility and the robust trainability provided by isolated quantum subsystems.
2. **Optimization:** We introduce a novel **Hybrid Quantum Forward-Forward** framework that fundamentally restructures QML training by replacing global backpropagation with a layer-wise, gradient-free optimization strategy. The core concept leverages the **Quantum Forward-Forward (QFF)** method to decompose deep quantum circuits into isolated, shallow layers, each optimizing a local "goodness" objective derived from separate positive (real) and negative (noise) data passes. Within each decoupled layer, we bypass classical optimizers by implementing the **Hybrid Quantum Genetic Algorithm (HQGA)**, which encodes circuit parameters into **quantum chromosomes**—superpositions representing multiple solution states simultaneously. By utilizing **entangled crossover** to stochastically correlate successful traits across the population and **quantum elitism** to reinforcement-learn optimal rotation angles, this approach allows for the training of complex, highly entangled models that are mathematically immune to the global barren plateau

problem while minimizing measurement overhead through local, efficient convergence.

3. **Temporal Expressibility:** Our **Quantum State Space Model (Q-SSM)** framework is built on a central hypothesis: **quantum circuits** can extract richer features from sequential data than classical networks by exploiting **entanglement and superposition**, while **classical gating mechanisms** remain optimal for **temporal memory management**. Our approach decomposes sequence modeling into two distinct functions: (1) feature extraction, where variational quantum circuits process input chunks, producing measurements that capture correlations across a 2^n-dimensional Hilbert space, and (2) temporal integration, where classical LSTM-style gates (forget, input, output) selectively accumulate information across chunks to model long-range dependencies. Three quantum branches are combined via trainable complex coefficients to enable interference-like effects, with an additional layer for bidirectionality through forward, backward, and global processing paths. This hybrid design assumes that the **expressivity advantage of quantum circuits lies in feature representation** rather than recurrence, and that near-term quantum hardware (or classical simulation) can practically execute shallow circuits while classical components handle the sequential state propagation that would otherwise require fault-tolerant quantum memory.
4. **Reliability:** We replace the synthetic Gaussian noise of classical diffusion models with the authentic, physical noise of quantum hardware. The core research approach assumes that hardware imperfections—such as decoherence and crosstalk—constitute a unique, learnable noise distribution that can be deterministically reversed. To bridge the incompatibility between discrete qubit outputs and continuous neural networks, the model utilizes **Fuzzy Quantum Logic (specifically POVMs)** to treat measurements as continuous "degrees of truth" rather than binary outcomes. By encoding data into quantum states, degrading them via natural hardware relaxation, and training a classical **Hardware-Attention U-Net** to invert this process, the system achieves a "hardware-native" architecture that exploits specific physical noise profiles for high-fidelity data generation and error mitigation. We certify their robustness using the **QUARK framework**.
5. **Validation:** We use **High-Complexity Domain Science** (HEP, Neuroscience, Cybersecurity) not just as applications, but as rigorous testbeds to certify the foundational advantages developed in Objectives 1-4.
   [High Energy Physics]
   We validate the foundational advantages of **Multi-Chip Ensemble QML**, specifically targeting the computational and precision bottlenecks inherent in processing massive particle collider data. The approach integrates hybrid quantum-classical models—including **Temporal Convolutional Network-VQC, Quantum Neural Networks (Q-ABCDisCo), and Quantum Vision Transformers** —with Multi-Chip Ensembles to demonstrate that quantum feature maps can capture complex particle correlations and solve ill-posed inverse problems with superior expressivity and parameter efficiency compared to classical deep learning. By successfully applying these algorithms to tasks like Higgs signal separation, background estimation, and waveform denoising , the project proves that Multi-Chip Ensemble QML offers a scalable solution for "Big Data" science, validating its ability to achieve high-precision results with significantly reduced computational resources and training data requirements.
   [Neuroscience]
   To decode the complex dynamics of the brain, our research utilizes **Quantum Transformers** and **Quantum State Space Models** for spatio-temporal analysis, employing Quantum Singular Value Transformation (QSVT) to reduce the computational cost of modeling long-range neural dependencies from quadratic to logarithmic scale. Complementing this, we frame brain connectivity as a quantum

optimization problem: **QAOA** structurally defines functional network boundaries by solving for optimal parcellation, while **VQE** dynamically simulates the brain's active inference process by minimizing a Hamiltonian representing Variational Free Energy to find Bayes-optimal posterior beliefs. Together, these methods exploit the high expressivity of quantum circuits to achieve robust generalization on small, high-dimensional neuroimaging datasets with minimal parameters.
[Cybersecurity]
We integrate the **Physics-Informed Fuzzy Quantum Diffusion Model** as a high-sensitivity anomaly detector with the **QUARK framework** for rigorous security benchmarking. We utilize the diffusion model to learn the unique physical noise fingerprint of a quantum processor, effectively establishing a hardware baseline that allows us to detect stealthy security breaches—such as hardware Trojans or induced crosstalk—by monitoring for specific deviations in the model's attention weights. Complementing this, we employ the QUARK framework to orchestrate attack simulations and validate system integrity, ensuring that our defenses are empirically verified against realistic, hardware-specific vulnerabilities.

Interdisciplinary Considerations & Stakeholder Knowledge

This proposal is inherently interdisciplinary, fusing three distinct scientific cultures to create a sum greater than its parts:

- **Quantum Computation & Domain Science (SNU, Yonsei):** Provides the rigorous understanding of quantum algorithm development (**Multi-chip Ensembles, Quantum State Space Models**), ansatz design (**Multi-Chip Ensembles**), and the domain knowledge of **Computational Neuroscience** and **High-Energy Physics**.
- **Computational Intelligence (Univ. Naples):** Injects mature classical methodologies—specifically **Fuzzy Set Theory** and **Bio-inspired Evolutionary Computing**—into the quantum domain to solve trainability issues that pure quantum theory has failed to address.
- **Application-focused QML & Robustness (Fraunhofer IKS):** Bridges the gap to industry by defining **quality benchmarks** and establishing robustness certifications. This stakeholder knowledge ensures our algorithms are not just theoretically interesting but reliable enough for safety-critical applications like cybersecurity.

Link to National and International Activities

The project leverages significant national ecosystems:

- **Korea (SNU/Yonsei):** Aligns with the national K-Quantum initiative. Prof. Yoo's involvement connects the project directly to the **CERN/LHC community**, allowing us to utilize international "Big Science" infrastructure as a testbed.
- **Germany (Fraunhofer IKS):** The work links to the **Munich Quantum Valley** ecosystem, focusing on industrial usage and certification standards for quantum software.
- **Italy (Univ. Naples):** Connects to the vibrant European community on soft computing and computational intelligence, bringing these classical AI strengths into the European Quantum Flagship landscape.

Methodology and Relevance to Objectives

Our methodology is divided into four integrated workstreams, directly addressing the Specific Objectives:

Methodology 1: Multi-Modal Multi-Chip Ensembles (Addressing Scalability) We employ **Optimal Quantum Architecture** to design orthogonal circuits for distinct data modalities (e.g., sMRI and fMRI) and **Optimal Quantum/Classical Connection** to strategically place entanglement only between globally dependent feature groups. This framework operates by partitioning high-dimensional multi-modal data across independent quantum chips—processing locally correlated inputs within isolated devices while using selective quantum connections to model necessary global dependencies—and integrating the outputs via classical ensemble learning.

- *Relevance:* This directly tackles **Objective 1**. It allows us to process multi-modal, high-dimensional datasets that exceed the capacity of a single chip, effectively addressing the connectivity and size limits of monolithic processors. Moreover, it overcomes the "curse of dimensionality" without requiring lossy classical dimension reduction, mitigates barren plateaus by preventing unnecessary global entanglement, and achieves superior noise resilience by simultaneously reducing both quantum error bias and variance, thereby providing a scalable and trainable solution compatible with emerging modular quantum hardware.

Methodology 2: Hybrid Physics-Informed Optimization (Addressing Trainability) Our approach replaces the prohibitive global backpropagation of deep circuits with a layer-wise training scheme (QFF), where each layer is optimized locally using a gradient-free Evolutionary Algorithm—**Hybrid Quantum Genetic Algorithms (HQGA)**—instead of gradient descent. This method works by restructuring the deep quantum circuit into locally trained layers using FF, where each layer's parameters are optimized not by gradient descent, but by the HQGA running directly on the quantum processor. In this workflow, the FF framework defines local "goodness" objectives (separating positive from negative data) for shallow sub-circuits, and the HQGA evolves a population of "quantum chromosomes"—superpositions of parameter configurations —using "entangled crossover" and "quantum elitism" to maximize these local objectives.

- *Relevance:* This addresses **Objective 2**. It resolves the "Trainability Trilemma" by simultaneously eliminating the **measurement overhead** of the parameter-shift rule (by being gradient-free) and mitigates the **Barren Plateau** problem through two distinct mechanisms: QFF's local objectives prevent exponential vanishing of gradients, while the HQGA's global search based on quantum mechanics to navigate the optimization landscape more efficiently than classical solvers, effectively permitting the training of **highly entangled, non-simulatable** quantum circuits that would otherwise be untrainable.

Methodology 3: Three-branch Superposition and LSTM-Gating for Q-SSM (Addressing Temporal Expressibility) We develop **Quantum State Space Models (Q-SSM)** by combining variational quantum circuits for feature extraction with classical LSTM-style gating for temporal memory management. The models process sequential data through chunked recurrence: each chunk is passed through three parallel quantum circuits whose measurement outputs are combined via trainable complex coefficients ($\alpha, \beta, \gamma \in \mathbb{C}$) inspired by quantum superposition—this "**measurement-based superposition**" enables interference-like effects while maintaining gradient flow. The resulting quantum features are then processed by classical forget, input, and output gates that selectively retain or discard information across chunks, with a final layer for bidirectional processing (forward, backward, and global branches).

- *Relevance:* This tackles **Objective 3**. This hybrid architecture is significant for three reasons: **(1) Expressivity:** the quantum circuits access a 2^n-dimensional Hilbert space with only O(n) parameters, providing a more expressive feature representation than equivalently-parameterized classical networks; **(2) Temporal modeling:** the

LSTM gating solves the critical limitation of prior quantum sequence models (which collapsed temporal information via pooling) by enabling selective memory; and **(3) Scalability:** the quantum circuit cost Q is constant with respect to sequence length L, so the overall O(L) complexity matches classical SSMs while potentially capturing non-classical correlations through entanglement that classical models cannot represent.

Methodology 4: Fuzzy-Quantum Diffusion & QUARK Certification (Addressing Reliability) The proposed Quantum Diffusion Model fundamentally redefines generative AI by **replacing the synthetic Gaussian noise of classical models with the authentic, physical noise inherent in quantum hardware**. In this hybrid architecture, a quantum processor serves as the "forward process," degrading clean data via natural error channels—such as amplitude damping, decoherence, or crosstalk—effectively turning hardware imperfections into a unique, physics-based feature. A classical deep neural network (e.g., a U-Net) is then trained to reverse this specific physical process, resulting in a model that learns the intricate "fingerprint" of the device, which can be used for generative tasks or bespoke error mitigation.
Fuzzy Quantum Logic acts as the essential mathematical bridge that makes this hybrid training possible. By utilizing Positive Operator-Valued Measures—the operational tool of fuzzy logic—we replace sharp, binary measurements (0 or 1) with continuous, "unsharp" values. This prevents the massive information loss that usually occurs during state collapse, allowing the classical network to train on "soft" probabilistic data that accurately reflects the texture of the quantum noise. This integration transforms the system from a discrete, binary model into a continuous diffusion process, enabling the neural network to interpret and correct quantum errors as fuzzy degrees of truth rather than simple bit-flips.

We will also implement the **QUARK** framework as a rigorous "Certification Layer" for our Fuzzy-Quantum Diffusion model. The implementation involves a three-step benchmarking protocol: (1) **Adversarial Perturbation:** We will subject the QML models (e.g., the Intrusion Detection System) to quantum-specific adversarial attacks. This involves applying unitary perturbations $U(\varepsilon)$ to the input quantum state $|\psi\rangle$ to simulate worst-case scenario shifts in the data manifold. (2) **Lipschitz Continuity Analysis:** Using QUARK's mathematical tools, we will compute the **local Lipschitz constant** of the quantum circuit. This metric provides a formal upper bound on how much the model's output can deviate given a bounded input perturbation. A lower Lipschitz constant mathematically certifies a higher degree of stability. (3) **Noise Stability Profiling:** We will inject calibrated NISQ noise models (depolarizing, dephasing) into the simulation and measure the degradation rate of the model's accuracy.

- *Relevance:* This addresses **Objective 4**. Standard diffusion fails when hardware noise mimics the diffusion noise; the hybrid architecture of the **Fuzzy-Quantum Diffusion** replaces the synthetic Gaussian noise of traditional AI with authentic physical noise, allowing for the creation of hardware-aware error mitigation tools and generative models that are robust to the inherent instability of NISQ devices. Moreover, by quantifying the **Lipschitz constant**, we convert the abstract concept of "robustness" into a precise, measurable engineering metric. This allows us to mathematically prove that the Fuzzy-Quantum integration successfully dampens the impact of input uncertainty, providing the "**Certified Robustness**" required for industrial trust.

Methodology 5: Domain-Specific Validation (Addressing Validation) We will rigorously preprocess the domain data: **CMS calorimeter images, high-level physics features, and detector waveform signals** (High Energy Physics), **fMRI/EEG time-series** (Neuroscience), and **Network Traffic logs** (Cybersecurity). **(1) High Energy Physics:** We integrate Multi-Chip Ensembles and QML algorithms to overcome the computational bottlenecks of high-energy physics data analysis. We use three hybrid quantum-classical models that integrate variational quantum circuits into established deep learning architectures—specifically replacing classical classifiers in the ABCDisCo background estimation model, output heads in Temporal Convolutional Networks (TCN-VQC) for denoising, and linear projection layers in Vision Transformers (ViT). We apply the **Multi-Chip Ensemble** framework by partitioning input data across independent quantum processors, thereby enabling complex feature extraction without the information loss typical of classical dimension reduction. Specifically, for **Q-ABCDisCo**, the ensemble distributes input kinematic variables across multiple chips to compute classifier scores, which are then optimized using **Distance Correlation penalties** to strictly enforce independence from background variables; for **TCN-VQC**, the framework splits time-series feature maps to reconstruct denoised waveforms via self-supervised **"Noise2Noise" strategies** that effectively train the model without requiring inaccessible clean labels ; and for **ClassicalQuantumViT**, the architecture partitions calorimeter image patch embeddings to facilitate parallelized, **quantum-enhanced attention mechanisms** essential for resolving fine-grained details in merged electron particle events. This distributed approach validates the utility of QML by demonstrating that complex, high-dimensional HEP algorithms can be successfully deployed on near-term hardware to achieve precise signal separation and noise rejection. **(2) Neuroscience:** To conduct spatio-temporal analysis of fMRI and EEG signals, we utilize **Quantum Transformers** and **Quantum State Space Models (Q-SSM)**, which employ Quantum Singular Value Transformation to reduce the computational complexity of attention mechanisms from quadratic to logarithmic scale, enabling the efficient modeling of long-term dependencies in neural dynamics. For brain connectivity research, we apply **QAOA** to solve the combinatorial problem of optimal network parcellation and structural mapping, while utilizing **VQE** to simulate the brain's functional inference processes by minimizing a Hamiltonian that represents variational free energy. **(3) Cybersecurity:** Our research utilizes the **Physics-Informed Fuzzy Quantum Diffusion Model** as a diagnostic sentinel and the **QUARK framework** as a rigorous benchmarking environment to fortify QML against hardware-level and adversarial threats. By deploying the Quantum Diffusion model, we establish a precise baseline of quantum processor behavior, allowing us to detect subtle anomalies—such as those introduced by hardware Trojans or induced crosstalk—through deviations in attention weights that signal potential integrity breaches. Concurrently, we leverage the QUARK framework's modular architecture to simulate specific attack scenarios (like noise injection) and standardize the evaluation of security metrics, ensuring that defenses are validated against realistic, hardware-aware noise profiles.

- *Relevance:* This methodology tackles **Objective 5**. It proves that the foundational breakthroughs translate into tangible advantages in accuracy, efficiency, and robustness for real-world scientific problems. Specifically, it addresses the critical limitations of classical machine learning regarding calculation volume and precision in the face of massive LHC datasets, providing a scalable, noise-resilient pathway to discover new physical phenomena and enhance standard model verification in the era of NISQ computing; it leverages the high expressivity of quantum circuits to achieve superior generalization on small, scarce neuroimaging datasets with fewer parameters than classical models, thereby overcoming the critical bottlenecks of overfitting and memory inefficiency in precision psychiatry; it moves beyond theoretical models to address the unique physical vulnerabilities of NISQ devices, providing a reproducible and hardware-specific methodology for certifying the security and reliability of quantum processors in critical applications.

Appropriateness of Methodology to Address Risks

Our methodology is specifically designed to narrow down options and mitigate high scientific risks:

1. **Risk: Barren Plateaus make training impossible.**

- *Mitigation:* We do not rely solely on one optimizer. By introducing **Evolutionary Algorithms**, we bypass gradient-dependence entirely. If QFF proves unstable, the Memetic Algorithm serves as a robust fallback global search method.

2. **Risk: Hardware noise destroys the Distributed Model's accuracy.**
   - *Mitigation:* **Ensemble methods** are inherently robust to noise. Since the Multi-Chip approach relies on aggregating independent models, the stochastic errors of individual chips tend to average out, acting as a form of algorithmic error mitigation without the cost of error correction codes.
3. **Risk: Hardware noise destroys generative fidelity.**
   - *Mitigation:* Instead of expensive error correction, we use **Fuzzy Logic** to "soften" the decision boundaries, making the diffusion model tolerant to the inherent uncertainty of NISQ devices.
4. **Risk: "Quantum Advantage" is elusive in real data.**
   - *Mitigation:* We diversify the validation across three distinct domains (HEP, Neuroscience, Cybersecurity). Each domain stresses a different quantum aspect (HEP=Dimensionality, Neuroscience=Temporal Correlation, Cybersecurity=Robustness). This portfolio approach maximizes the probability of demonstrating a clear foundational advantage in at least one key area certified by the **QUARK framework**.

1.4 Interdisciplinary nature

Describe the research disciplines involved and the range of added value from interdisciplinarity, including measures for exchange, cross-fertilisation and synergy.

Research Disciplines Involved

This proposal is built upon a radical convergence of distinct scientific cultures. We move beyond standard multidisciplinary collaboration—where groups work in parallel—to true **interdisciplinarity**, where methods from one field are mathematically integrated into the core protocols of another.

1. **Quantum Information & QML Architectures:** This discipline provides the fundamental "operating system" for the project—entanglement, superposition, and Hamiltonian dynamics—and the theoretical framework for the **Multi-Chip Ensembles** architecture.
2. **Computational Intelligence & Soft Computing:** This discipline injects mature classical methodologies—specifically **Fuzzy Logic** and **Evolutionary/Memetic Algorithms**—into the quantum stack. This provides the mathematical tools to handle uncertainty and non-convex optimization, areas where pure quantum theory currently struggles.
3. **Quantum Software Engineering & Reliability:** This discipline bridges the gap between theory and industry. It provides the **QUARK Framework** (Quantum Computing for Artificial Intelligence and Reliable Knowledge), which serves as the rigorous standard for benchmarking, certifying, and ensuring the robustness of the developed algorithms.
4. **Domain Sciences (Neuroscience, High Energy Physics, Cybersecurity):** These disciplines provide the complex, high-dimensional "reality" against which the quantum algorithms are tested. They serve as rigorous validation environments for foundational claims regarding temporal correlation (Neuroscience), massive scalability (HEP), and adversarial robustness (Cybersecurity).

Range of Added Value from Interdisciplinarity

The primary added value is the creation of a **"Hybrid Robustness"** that cannot be achieved by physics or computer science alone. We leverage specific domain characteristics to solve foundational quantum challenges:

- **Neuroscience x Fuzzy Quantum Logic (The "Bio-Fuzzy" Synergy):**
  Biological signals (e.g., EEG/fMRI) are inherently non-stationary and noisy, often confounding standard quantum algorithms. By integrating Fuzzy Set Theory with Quantum Diffusion Models, we create a unique synergy: "Fuzziness" models the biological variability and noise, while "Quantum Diffusion" captures the complex, non-local temporal correlations in brain activity. This cross-fertilization creates a new class of generative models capable of synthesizing high-fidelity biological data, a feat unattainable by classical logic or pure quantum mechanics alone.
- **Cybersecurity x Evolutionary QML (The "Adversarial" Synergy):**
  In cybersecurity, network intrusion patterns change dynamically as attackers evolve. Gradient-based Quantum Machine Learning (QML) is often brittle against such adversarial shifts. By fusing Cybersecurity principles with Evolutionary and Memetic Algorithms, we develop QML models that do not merely "learn" a static dataset but "evolve" to find robust solutions in a dynamic threat landscape. We further enhance this by applying the QUARK Framework to certify the model's robustness against adversarial perturbations, transforming "Quantum Advantage" into "Certified Reliability".
- **High Energy Physics x Distributed Computing (The "Scale" Synergy):**
  The validation of distributed quantum computing requires data with intricate, non-local entanglement structures. High Energy Physics (LHC) data provides exactly this complexity. By applying Multi-Chip Ensemble protocols to particle tracking problems, we validate the "scalability resource theory" using real-world physical interactions. We prove that parallel quantum processors, working in concert via ensemble aggregation, can reconstruct the global correlations of subatomic particles without requiring a single monolithic fault-tolerant device.

Measures for Exchange, Cross-fertilisation, and Synergy

To ensure this interdisciplinary fusion occurs effectively, we have designed specific mechanisms for knowledge transfer:

1. **Cross-Disciplinary "Methodology Swaps":**
   We will organize bi-annual workshops where domain experts teach their core tools to the theory groups. For example, experts in Computational Intelligence will train the physicists on Fuzzy Logic Controller design, while QML researchers will train the domain scientists on the Multi-Chip Ensemble ansatz and Q-SSM. This ensures that "Fuzzy-Quantum" integration occurs at the code level, not just the conceptual level.
2. **Joint "Challenge" Sprints:**
   Research activities will be organized around "Challenge Datasets" provided by the domain experts.
   - **The Neuro-Challenge:** The theory group must apply **Fuzzy-Diffusion** to synthesize realistic EEG/fMRI data provided by the Neuroscience experts.
   - **The Cyber-Challenge:** The optimization group must use Evolutionary QML to detect anomalies in Network Traffic logs provided by the Cybersecurity experts, validated against QUARK robustness metrics.
     These sprints force the theoretical algorithms to adapt to the rigorous constraints of real-world domain data.
3. **Unified "Physics-Aware" Software Repository:**
   All developed algorithms (e.g., Fuzzy-Diffusion, Multi-Chip Ensembles) will be integrated into a shared, modular software repository. This enforces technical

cross-fertilization, as the "Physics" modules (e.g., Hamiltonians) must interface correctly with the "Logic" modules (e.g., Fuzzy Controllers) and the QUARK benchmarking tools, ensuring the final deliverable is a cohesive library rather than isolated scripts.

1.5 Gender dimension and open science practices

Describe how gender dimension in research content is addressed.
Describe how open science practices including sharing and management of research outputs are implemented.

Gender Dimension in Research Content

While the foundational quantum physics and abstract optimization algorithms (Goal 1) are gender-neutral by nature, the **Neuroscience application (Goal 2.1)** involves the processing of human biological data, where the gender dimension is scientifically critical.

- **Algorithmic Bias Mitigation in Neuroscience:** Research indicates that EEG/fMRI signals and brain-computer interface (BCI) performance can exhibit sex-specific variabilities due to anatomical and physiological differences. A QML model trained predominantly on male subjects may fail to generalize to female subjects, creating a biased algorithmic standard.
- **Implementation:** In the **Fuzzy-Quantum Diffusion** workstream, we will implement a "Gender-Stratified Validation" protocol. We will ensure that the training datasets for EEG/fMRI synthesis are balanced (50/50 male/female). Furthermore, we will treat "biological sex" as a distinct fuzzy variable within the Fuzzy Logic Controller, allowing the model to explicitly learn and adapt to sex-specific signal features rather than treating them as noise. This ensures the developed **Robust QML** tools are universally applicable and free from biological bias.

For the remaining research lines (HEP, Cybersecurity, and Foundational QML), the content is gender-neutral. However, we proactively address the gender dimension in the **research process** by ensuring gender-balanced project management and recruitment, aligning with the **QuantERA Gender Equality Statement**. The consortium includes female leadership (Prof. Jeanette Lorenz, Fraunhofer IKS) in a key Principal Investigator role, serving as a role model for early-career researchers.

Open Science Practices

We strictly adhere to the **Horizon Europe Open Science policy**, adopting a "As Open As Possible, As Closed As Necessary" approach to maximize the impact and reproducibility of our foundational breakthroughs.

1. Open Access to Publications

- **Policy:** We commit to **100% Open Access** for all peer-reviewed scientific publications resulting from this project.
- **Implementation:** We will adopt the "Green Road" by depositing the final peer-reviewed manuscripts or preprints in open repositories (e.g., **arXiv**, **Zenodo**) immediately upon acceptance, ensuring compliance with national funder mandates (e.g., NRF Korea, DFG). Where budget allows, we will opt for "Gold Open Access" in high-impact journals.

2. Research Data Management (FAIR Principles)

We will implement a Data Management Plan (DMP) within the first 3 months, adhering to the FAIR principles (Findable, Accessible, Interoperable, Reusable).

- **Findable & Accessible:** All non-sensitive simulation data, trained QML model weights, and benchmark results (e.g., from the HEP and Optimization tasks) will be deposited in **Zenodo** with a unique Digital Object Identifier (DOI).
- **Interoperable:** Data will be stored in standard, open formats (e.g., HDF5 for large datasets, JSON for model hyperparameters) rather than proprietary formats.
- **Reusable:** All datasets will be released under **Creative Commons licenses (CC BY)** to encourage reuse by the global quantum community.

3. Open Source Software

The "Physics-Aware QML" algorithms (QFF, Fuzzy-Diffusion) are the core output of this project.

- **Repository:** We will establish a public **GitHub repository** for the project.
- **Licensing:** All code will be released under a permissive open-source license (e.g., **MIT** or **Apache 2.0**) to foster industrial adoption and academic collaboration.
- **Documentation:** We will provide Jupyter notebook tutorials demonstrating how to apply our distributed and evolutionary QML protocols to standard datasets, lowering the barrier to entry for other researchers.

4. Management of Sensitive Data

For Cybersecurity (Goal 2.3) and Neuroscience (Goal 2.1), data privacy is paramount.

- **Anonymization:** Any human EEG/fMRI data will be strictly pseudonymized before processing.
- **Security:** Cybersecurity datasets containing potentially sensitive network vulnerability patterns will be shared only within the consortium via secure, encrypted channels, following the "As Closed As Necessary" principle to prevent misuse.

2. Impact (max. 3 pages; Weight: 25%)

2.1 Expected impacts

Be specific, and provide only information that applies to the proposal and its objectives. Wherever possible, use quantified indicators and targets.
Describe how the project will contribute to the expected impacts (see ‘Research Targeted in the Call’ of the Call Announcement).
Describe the importance of the technological outcome with regard to its transformational impact on technology and/or society.

Contribution to Expected Impacts

This project directly addresses the "Quantum Phenomena and Resources" (QPR) call objective to *"explore novel quantum phenomena, concepts, resources... and address major challenges that prevent broad applications."* We contribute to this by establishing a **Resource Theory for Scalable and Reliable QML**, impacting the field as follows:

1. Deepening the Understanding of Quantum Resources (Scalability via Ensembles)

- **Impact:** We will establish **"Parallel Quantum Processing"** as a quantifiable resource. By validating the **Multi-Chip Ensemble** architecture, we will determine the precise scaling laws between *ensemble size* (number of parallel chips) and *model expressibility*. This moves the field away from the dependency on single, large-volume entanglement towards **"Collective Quantum Advantage"** derived from aggregating diverse, smaller quantum models.
- **Target:** Demonstrate that a **Multi-Chip Ensemble** (using architectures like Quantum Transformers) can achieve **>90%** of the accuracy of a monolithic ideal circuit while reducing the qubit requirement per chip by **50%**. This proves that quantum advantage is accessible on networked NISQ devices without requiring full inter-chip connectivity or complex teleportation protocols.

2. Enhancing Robustness in the Presence of Decoherence

- **Impact:** We will fundamentally transform how noise is handled in generative QML. By shifting from resource-heavy "Error Correction" to **"Fuzzy Uncertainty Management,"** we enable quantum generative models to function on current noisy hardware. Crucially, we will use the **QUARK framework** to rigorously quantify this robustness.
- **Target:** Achieve a **2x improvement in Fréchet Inception Distance (FID)** scores for quantum-generated data (e.g., EEG/fMRI, particle jets) compared to standard Quantum GANs under realistic NISQ noise levels (10^{-3} gate error rates). Furthermore, we aim to achieve **certified robustness** (measured by QUARK metrics like Lipschitz continuity) against input perturbations.

3. Identifying New Opportunities and Applications (The Validation Feedback Loop)

- **Impact:** We will provide the first rigorous proof-of-concept that Foundational QML can solve "Big Science" problems that are intractable for classical methods due to data complexity.
- **Target:**
  - **High Energy Physics:** Demonstrate a **>5% improvement** in signal-to-background ratio for LHC particle tracking compared to classical Deep Learning, using Multi-Chip Ensembles to handle the high dimensionality of jet data.
  - **Neuroscience:** Demonstrate that **Fuzzy-Quantum** models can capture **long-range temporal correlations** in EEG/fMRI data that classical RNNs/Transformers miss, offering new biomarkers for neurological conditions.

Transformational Impact on Technology and Society

The technological outcome of this project—a **"Physics-Aware" QML Stack**—will have a transformational impact beyond the academic quantum community:

1. Technological Sovereignty & Standardization (The "Trust" Impact)

- **Context:** Currently, there is no standard for certifying that a quantum algorithm is "safe" or "robust."
- **Our Contribution:** Led by **Fraunhofer IKS**, we will leverage the **QUARK framework** to define the first **"Quantum Software Reliability Standard"** for QML. This includes specific metrics for robustness against adversarial attacks (validated via the Cybersecurity use case).

- **Society:** This is a prerequisite for deploying quantum AI in safety-critical infrastructure (energy grids, autonomous vehicles, banking). By establishing these certification protocols, we pave the way for the European and Global industry to adopt quantum solutions with confidence.

2. Democratization of Quantum Access (The "Scale" Impact)

- **Context:** "Quantum Advantage" is currently restricted to elite labs with the largest chips (100+ qubits).
- **Our Contribution:** The **Multi-Chip Ensemble** protocol allows researchers to pool multiple smaller, cheaper quantum chips (e.g., 20-qubit devices) to solve larger problems via classical aggregation.
- **Society:** This democratizes access to high-performance quantum computing. Universities and smaller companies with modest hardware can collaborate to run "Virtual Large-Scale" experiments, accelerating the global pace of quantum innovation.

3. Advanced Data Privacy in Healthcare (The "Fuzzy-Quantum" Impact)

- **Context:** Sharing medical data (like EEG/fMRI) is difficult due to privacy regulations (GDPR).
- **Our Contribution:** Our **Fuzzy-Quantum Diffusion** model can generate high-fidelity *synthetic* medical data that preserves the statistical properties of the original patients without revealing their identities.
- **Society:** This unlocks massive datasets for medical research, allowing hospitals to share "Quantum-Synthetic Data" for drug discovery or disease modeling without compromising patient privacy.

2.2 Dissemination, exploitation of results, communication

Provide a plan for disseminating and exploiting the project results beyond the project itself (scientifically and/or economically).
Results include any data produced in the framework of the project. If applicable, describe how data curation and distribution can be ensured beyond the project duration.

When applicable, describe how subsequent translation of research results into innovations and development of new products and/or services are facilitated

Describe the proposed communication measures for promoting the project and its findings during the period of the project. Measures should be with clear objectives. They should be tailored to the needs of different target audiences, including groups beyond the project's own community.
Where relevant, include measures for public/societal engagement on issues related to the project.

Dissemination of Results (Scientific & Open Science)

We adopt a "Physics-to-Product" dissemination strategy, ensuring that our foundational breakthroughs are rigorous enough for high-impact physics journals yet accessible enough for software engineering adoption.

1. Scientific Publications (100% Open Access):`

- **Target:** We plan for **12+ high-impact peer-reviewed publications** in leading venues.
    - *Nature Physics / Physical Review Letters:* For the foundational **Multi-Chip Ensembles** and **Quantum Time-Series Transformer** results.
    - *IEEE Transactions on Evolutionary Computation:* For the **Memetic QML** and **Quantum Forward-Forward** algorithms.
    - *Quantum Science and Technology:* For the **QUARK-certified reliability benchmarks** and domain validations (HEP, Neuro).
- **Mechanism:** All preprints will be uploaded to **arXiv** (quantum-ph) immediately. We will cover Gold Open Access fees where possible to ensure immediate accessibility.

2. Open Source "Physics-Aware QML" Library:

- **Deliverable:** The core algorithms—**Multi-Chip Ensembles** (Scalability), **QFF-Evo** (Optimization), and **Fuzzy-Diffusion** (Robustness)—will be released as a modular Python library. Crucially, this library will include a **QUARK-connector** module, allowing users to automatically benchmark their models against the robustness metrics defined by Fraunhofer IKS.
- **Curation:** The code will be hosted on **GitHub** with a permissive license (Apache 2.0).
- **Documentation:** We will provide comprehensive Jupyter notebooks demonstrating how to replicate our results on the specific "Challenge Datasets" (LHC , EEG/fMRI, Cyber-logs), lowering the barrier for other researchers to verify our claims.

3. Data Curation & Distribution:

- **Repositories:** The **Zenodo** platform will be used to host the non-sensitive "Challenge Datasets" (e.g., the anonymized EEG/fMRI benchmarks and specific LHC subsets). Each dataset will be assigned a DOI.
- **Long-term Preservation:** Metadata will follow the **FAIR** (Findable, Accessible, Interoperable, Reusable) principles. The data management plan (DMP) ensures these assets remain available for at least 10 years.

Exploitation of Results (Innovation & Standardization)

Our exploitation strategy focuses on transforming "Foundational Resources" into "Reliable Tools" for the quantum ecosystem.

1. Standardization of Quantum Software Reliability (Led by Fraunhofer IKS):

- **Innovation:** We will integrate the project's **Fuzzy-Quantum** robustness metrics into the industrial-grade **QUARK Framework**.
- **Facilitation:** Fraunhofer IKS will leverage its position in the **Munich Quantum Valley** and industrial networks to propose the **QUARK framework** as a preliminary industrial standard for certifying safety-critical QML (e.g., for automotive or finance sectors). This facilitates the translation of our research into a marketable "Certification Service."

2. Scalable "Virtual" Quantum Computing (Led by SNU/Yonsei):

- **Innovation:** The **Multi-Chip Ensemble** protocol enables users with small, fragmented chips to solve larger problems by aggregating parallel quantum models (e.g., Quantum Transformers).
- **Exploitation:** We will explore licensing this "Ensemble Strategy" to cloud quantum providers. It serves as an algorithmic optimization layer that can run on top of existing

NISQ hardware, instantly upgrading the capability of current machines without hardware changes.

3. Domain-Specific AI Tools (Led by Naples/SNU):

- **Innovation:** The **Neuro-Quantum Generative Model** addresses the scarcity of medical data using Fuzzy-Diffusion.
- **Exploitation:** This tool can be exploited by hospitals or MedTech startups to generate privacy-preserving synthetic patient data (EEG/fMRI), solving a major bottleneck in medical AI development while complying with GDPR.

Communication Measures (Public & Community Engagement)

We will promote the project to diverse audiences with tailored messages.

1. Academic & Developer Community (The "Users"):

- **Measure:** Organize a **"Quantum Robustness Hackathon"** (Year 3).
- **Objective:** Invite students and developers to try and "break" our Robust QML models using **QUARK's adversarial attack toolkits** or use our **Multi-Chip protocols** to solve a distributed learning challenge.
- **Target:** 50+ participants, fostering a community of early adopters for our software tools.

2. General Public (The "Society"):

- **Measure:** "Quantum for Big Science" Webinar Series.
- **Objective:** Explain *why* we need quantum computers for things like the LHC (searching for new particles) or Brain Science, demystifying the "hype" and highlighting the physics of entanglement.
- **Target:** Reach 500+ viewers via YouTube/University channels. We will specifically highlight the role of **Fuzzy Logic** as a bridge between human reasoning and quantum mechanics, making the concepts more relatable.

3. Policymakers & Industry (The "Funders"):

- **Measure:** Policy Brief on "Standardizing Quantum AI."
- **Objective:** Produced by Fraunhofer IKS, this brief will explain the necessity of **Robustness Certification (QUARK)** before QML is deployed in critical infrastructure, advocating for continued funding in reliability research (QPR topic).

3. Implementation (Weight: 25%)

3.1 Work Plan (max. 2 pages)

Provide a brief presentation of the overall structure of the work plan.
Clearly define the intermediate targets.
Provide a timing of the different work packages and their components (Gantt chart or similar).
Provide a graphical representation of the work packages components showing how they inter-relate (Pert chart or similar).

Overall Structure of the Work Plan

The project is organized into **six interconnected Work Packages (WPs)** designed to create a feedback loop between theoretical development and empirical validation.

- **WP1: Management & Coordination (Lead: SNU)** ensures scientific alignment, risk management, and administrative compliance across the consortium.
- **WP2: Scalable Distributed Quantum Architectures (Lead: SNU)** focuses on **Goal 1.1**. It develops the **Multi-Chip Ensemble** protocols to enable scalable QML on fragmented hardware by aggregating parallel quantum models.
- **WP3: Physics-Informed Optimization & Generative Models (Lead: Univ. Naples)** focuses on **Goals 1.2 & 1.3**. It develops the **Quantum Forward-Forward (QFF)** algorithm, **Memetic/Evolutionary strategies**, and **Fuzzy-Quantum Diffusion** models to solve trainability and noise challenges.
- **WP4: High-Complexity Domain Validation (HEP & Neuroscience) (Lead: Yonsei)** focuses on **Goals 2.1 & 2.2**. It prepares the "Big Science" datasets (LHC, EEG/fMRI) and benchmarks the algorithms from WP2/3 against classical SOTA to prove quantum advantage.
- **WP5: Reliability, Safety & Certification (Lead: Fraunhofer IKS)** focuses on **Goal 2.3** and the **Reliability Standard**. It benchmarks the robustness of the developed models against adversarial attacks (Safety & Cybersecurity) and defines certification metrics using the **QUARK framework**.
- **WP6: Dissemination, Exploitation & Communication (Lead: Fraunhofer IKS)** ensures Open Science compliance, code release, and industrial engagement.

Intermediate Targets

We have defined four critical Intermediate Targets (ITs) to monitor progress:

- **IT-1 (Month 12): Theoretical Foundations Established.**
  - Mathematical formulation of the **Multi-Chip Ensemble** ansatz complete.
  - Fuzzy Logic Controllers for Quantum Diffusion defined.
  - Domain datasets (LHC, EEG, Network Logs) preprocessed and feature-mapped to Hilbert space.
- **IT-2 (Month 24): Algorithmic Validation (Simulation).**
  - QFF-Evolutionary algorithm demonstrates convergence on >10 qubit simulations where Backpropagation fails (Barren Plateau proof).
  - Fuzzy-Diffusion generates synthetic EEG data with statistical fidelity comparable to classical GANs.
- **IT-3 (Month 30): Cross-Domain Integration.**
  - **Multi-Chip Ensemble** protocol successfully reconstructs correlations in LHC particle jets by aggregating outputs from 2+ parallel chips.
  - Cybersecurity models pass the preliminary robustness certification against gradient-based attacks using **QUARK metrics**.
- **IT-4 (Month 36): Final "Physics-Aware" Library.**
  - Release of the integrated open-source library.
  - Publication of final benchmarks demonstrating advantage in scalability (WP2) and robustness (WP5).

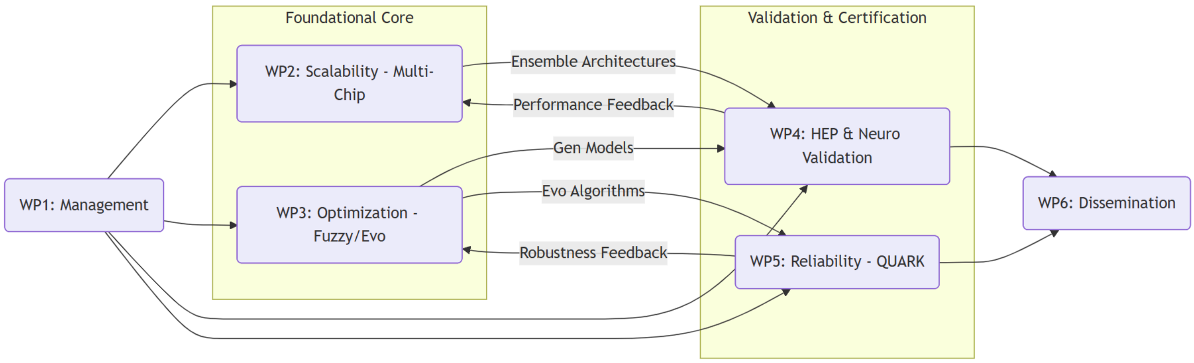
Timing of Work Packages

Work Package	M1-6	M7-12	M13-18	M19-24	M25-30	M31-36
WP1: Management	**====**	**====**	**====**	**====**	**====**	**====**
WP2: Scalable Arch (Multi-Chip)	**Definition of Protocols/ Architectures**	**Algorithm Development**	**Simulation-based Proof of Concept**	**Optimization & Refinement**	**Integration into Unified Library**	**Reporting (Deliverables/Periodic Reports)**
WP3: Opt & Gen Models (Fuzzy)	**Definition of Protocols/ Architectures**	**Algorithm Development**	**Simulation-based Proof of Concept**	**Optimization & Refinement**	**Integration into Unified Library**	**Reporting (Deliverables/Periodic Reports)**
WP4: Domain Validation (HEP/Neuro)	**Data Collection & Preprocessing**	**Quantum Feature Mapping**	**Initial Testing & Debugging**	**Benchmarking against Classical SOTA**	**Final Domain Validation**	**Reporting (Deliverables/Periodic Reports)**
WP5: Reliability (QUARK) & Cyber	**Requirements**	**Metric Definition (Robustness/Fuzziness)**	**Initial Testing & Debugging**	**Adversarial Attack Testing**	**Reliability Certification**	**Reporting (Deliverables/Periodic Reports)**

WP6: Dissemination	**Dissemination Planning**	**---**	**Scientific Publication**	**---**	**Hackathon Event**	**Final Conference Presentation**

The workflow follows a clear dependency path where **Foundational Theory (WP2/3)** feeds into **Domain Validation (WP4/5)**, with continuous feedback loops for refinement.

1. **Input:** Domain Requirements (from **WP4/WP5**) → define the constraints for **WP2** (Scalability needs for LHC) and **WP3** (Robustness needs for Neuro/Cyber).
2. **Development: WP2** (Multi-Chip Ensembles) and **WP3** (Fuzzy/Evo) develop the core algorithms in parallel.
3. **Integration:** The algorithms are integrated into a unified framework.
4. **Validation:**
   - **WP4** takes the **WP2** Multi-Chip engine to process LHC data.
   - **WP4** takes the **WP3** Fuzzy-Diffusion model to synthesize EEG/fMRI data.
   - **WP5** takes the **WP3** Evolutionary models and subjects them to Adversarial Attacks (Cybersecurity) to certify reliability using the **QUARK framework**.
5. **Output:** Validated results flow into **WP6** for dissemination.



3.2 Work Packages

(max. 1 page per WP)

For the description of each work package, please use the template provided. Use as many templates as needed.

WP 1	**Project Management & Coordination**						Start Month: 1	End month : 36
Contribution of project partners								
Partner number[1]	**1**	2	3	4	5	6	7	8
Total effort per partner (person*months)								

[1] ***Bold*** *the partner number of the work package leader*

Aim of the WP	
To ensure efficient administrative and scientific coordination, timely reporting to national funding agencies and the QuantERA secretariat, and effective risk/data management.	
Tasks	
T1.1	**Consortium Coordination & Monitoring (M1–M36: SNU; All)**[2] Day-to-day management of the project; organization of bi-annual consortium meetings (virtual/physical); monitoring of milestones and KPIs.
T1.2	**Risk & Innovation Management (M1–M36: SNU; Fraunhofer)** **Task title (start month – end month: responsible partner; involved partners)** Continuous monitoring of scientific and technical risks; execution of mitigation plans; management of Intellectual Property Rights (IPR).
T1.3	**Data Management & Ethics (M1–M36: SNU; Fraunhofer)** Implementation of the Data Management Plan (DMP) ensuring FAIR principles; oversight of ethical compliance regarding EEG/fMRI data (Neuroscience) and sensitive network logs (Cybersecurity).

Deliverable	**Month of delivery**	**Title of deliverable**
D1.1	M3	Data Management Plan
D1.2	M18	Mid-term Progress Report
D1.3	M36	Final Project Report

WP 2	**Scalable Distributed Quantum Architectures**							Start Month: 1	End Month : 36
Contribution of project partners									
Partner number	1	2	3	4	5	6	7	8	
Total effort per partner (person*months)									

Aim of the WP	
To develop theoretical protocols for "Virtual Large-Scale" quantum computing by implementing Multi-Chip Ensembles, enabling scalable QML on fragmented NISQ hardware via parallel model aggregation.	
Tasks	
T2.1	**Multi-Chip Protocol Definition (M1–M12: SNU; Naples, Fraunhofer)**

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[2] *For instance: T1.1 Development of something (M3-M6; responsible: 3; involved: 1, 4)*

	Mathematical formulation of ensemble learning strategies for distributed QPUs; development of aggregation logic (voting/averaging) to combine outputs from independent quantum models.
T2.2	**Multi-chip Architecture Optimization (M7–M24: SNU; Naples, Fraunhofer)** Design of scalable QML architectures (Quantum Transformers, HQTCN) optimized for ensemble deployment, targeting improved generalization and data-efficiency; with a focus on minimization of qubits-per-chip requirements.
T2.3	**Integration & Testing (M13–M36: SNU; Naples, Fraunhofer)** Implementation of the Multi-Chip protocols into a unified software module; verification of scalability, trainability, performance and fidelity retention across simulated distributed environments.

Deliverable	**Month of delivery**	**Title of deliverable**
D2.1	M12	Report on Multi-Chip QUARK Protocols
D2.2	M24	Software Module for Distributed Circuit Cutting

WP 3	**Physics-Informed Optimization & Generative Models**					Start month: 1	End month : 36
Contribution of project partners							

Partner number	1	2	3	4	5	6	7	8
Total effort per partner (person*months)								

Aim of the WP

To solve the "Trainability" and "Noise" bottlenecks by developing the Quantum Forward-Forward algorithm (QFF) driven by Memetic Evolution, and Fuzzy-controlled Quantum Diffusion models.

Tasks	
T3.1	**Evolutionary QFF Development (M1–M18: Naples; SNU)** Development of Memetic Algorithms (MA) to drive Quantum Forward-Forward learning; benchmarking convergence against Backpropagation in Barren Plateau landscapes.
T3.2	**Fuzzy Logic Controller Design (M7–M24: Naples; SNU)** Design of Fuzzy Logic Controllers (FLC) to model NISQ noise as uncertainty; integration of FLC into the reverse diffusion process of generative models.

T3.3	**Robust Model Implementation (M19–M36: Naples; SNU, Fraunhofer)** Full implementation of the Fuzzy-Quantum Diffusion model with built-in noise resilience.

Deliverable	**Month of delivery**	**Title of deliverable**
D3.1	M18	Algorithm Definition: QFF-Evolutionary & Fuzzy-Diffusion
D3.2	M30	Benchmarking Report: Trainability & Noise Resilience

WP 4	**High-Complexity Domain Validation (HEP & Neuro)**						Start month	End month
Contribution of project partners								
Partner number	1	2	3	4	5	6	7	8
Total effort per partner (person*months)								

Aim of the WP

To validate the foundational breakthroughs (WP2/WP3) using massive, complex datasets from High Energy Physics (LHC) and Neuroscience (EEG, fMRI), proving quantum advantage in scalability and temporal modeling.

Tasks

T4.1	**Domain Data Preprocessing (M1–M6: Yonsei; SNU, Fraunhofer)** Preparation of LHC particle jet data (Yonsei) and EEG/fMRI time-series (SNU/Naples); implementation of quantum feature maps (Fraunhofer).
T4.2	**HEP Scalability Validation (M13–M30: Yonsei; Naples, Fraunhofer)** Application of Multi-Chip Ensembles (from WP2) to LHC jet classification; benchmarking accuracy and generalization vs. single-chip and classical SOTA baselines.
T4.3	**Neuro-Generative Validation (M13–M30: SNU; Naples, Fraunhofer)** Application of Fuzzy-Diffusion (from WP3) to synthesize EEG/fMRI data; evaluation of statistical fidelity (FID score) and privacy preservation.

Deliverable	**Month of delivery**	**Title of deliverable**
D4.1	M6	Curated "Challenge Datasets" (LHC & EEG/fMRI)
D4.2	M36	Validation Report: Roadmap to Quantum Advantage in HEP & Neuro

WP 5	Reliability, Safety & Certification						Start month: 1	End month : 36
Contribution of project partners								
Partner number	1	2	3	4	5	6	7	8
Total effort per partner (person*months)								

Aim of the WP

To define the first "QML Reliability Standard" and benchmark and certify the robustness of the developed QML models against cybersecurity threads and privacy attacks.

Tasks

T5.1	**Reliability Metrics and Safeguard techniques (M1–M12: Fraunhofer; All)** Definition of formal metrics for QML robustness (e.g., Lipschitz continuity, noise stability, privacy budget, data leakage)
T5.2	**Cybersecurity Adversarial Testing (M13–M30: Fraunhofer; Naples)** Application of Evolutionary QML to Network Intrusion Detection; stress-testing the models with gradient-based adversarial attacks to measure robustness.
T5.3	**Certification Framework (M25–M36: Fraunhofer)** Synthesis of results into a formal QUARK Certification protocol for industrial QML applications.

Deliverable	**Month of delivery**	**Title of deliverable**
D5.1	M18	Robustness Report:
D5.2	M36	QML Reliability Standard, e.g. certification protocols

WP 6	Dissemination, Exploitation & Communication						Start month: 1	End month : 36
Contribution of project partners								
Partner number	1	2	3	4	5	6	7	8
Total effort per partner (person*months)								

Aim of the WP

To maximize impact through Open Science publications, release of the "Physics-Aware QML" library, and engagement with the industrial and public sectors.

Tasks	
T6.1	**Scientific Dissemination & Open Science (M1–M36: All)** Publication of results in Open Access journals; management of the GitHub repository and Zenodo data archives.
T6.2	**Exploitation & Standardization (M13–M36: Fraunhofer; All)** Promotion of QUARK Reliability Standards to industrial bodies (e.g., Munich Quantum Valley); exploration of licensing models for the Multi-Chip middleware.
T6.3	**Communication & Events (M1–M36: Fraunhofer; SNU)** Organization of the "Quantum Robustness Hackathon" (M30); public webinars on "Quantum for Big Science"; website maintenance.

Deliverable	**Month of delivery**	**Title of deliverable**
D6.1	M36	Open Source "Physics-Aware QML" Library Release
D6.2	M36	Final Plan for Exploitation and Standardization

Work package overview (total effort per WP and partner in person.months)

Use as many lines and columns as needed.

Partner	WP1	WP2	WP3	WP4	WP5	WP6	Total
1							
2							
3							
4							
5							
6							
Total							

3.3 Management structure, milestones, risk assessment

(max. 2 pages)

Describe the organisational structure and the decision-making including a list of milestones (template provided). A milestone is a major and visible achievement. It should be SMART: Specific, Measurable, Attainable, Relevant, Time-bound.
Explain why the organisational structure and decision-making mechanisms are appropriate to the complexity and scale of the project.

Organisational structure and decision-making

The project management structure is designed to ensure efficient coordination between the Asian (Korea) and European (Germany, Italy) partners, facilitating the rigorous feedback loop between foundational theory and domain validation.

- **Project Coordinator (PC):** Prof. Jiook Cha (SNU, Partner 1) is the single point of contact for the QuantERA Secretariat. The PC is responsible for overall progress monitoring, submission of deliverables, and conflict resolution.
- **General Assembly (GA):** The ultimate decision-making body, composed of one representative from each partner (Cha, Yoo, Acampora, Lorenz). The GA meets bi-annually to review progress against Milestones. Decisions are made by consensus; if consensus fails, a majority vote (one vote per partner) applies.
- **Scientific Steering Committee (SSC):** Composed of the Work Package Leaders. The SSC meets quarterly to ensure technical alignment (e.g., ensuring WP2's Multi-Chip Ensemble protocols meet WP4's HEP data requirements).
- **Innovation & Ethics Board:** Led by **Fraunhofer IKS (Partner 4)**, this board oversees IPR management, open science compliance via the QUARK framework standards, and ethical handling of Neuroscience/Cybersecurity data.

This structure is appropriate because it is **lean** (minimizing administrative overhead for a 4-partner consortium) yet **robust** (ensuring all scientific disciplines—Physics, CS, Engineering—have equal voting power in the GA).

List of milestones

Use as many lines as needed but try to limit the number of milestones.

Milestone	Delivery month	WP involved	Title
M1	M12	WP2, WP5	**Foundational Architectures Defined:** Multi-Chip Ensemble protocol specs and QUARK Reliability Metrics defined.
M2	M18	WP3	**Algorithmic Proof-of-Concept:** QFF-Evolutionary algorithm demonstrates convergence in simulation (Barren Plateau avoidance).
M3	M30	WP4, WP5	**Cross-Domain Validation:** Successful application of Multi-Chip Ensembles to LHC data and Fuzzy-Diffusion to EEG data.

M4	M36	WP6	**Final Release:** "Physics-Aware QML" Open Source Library released with QUARK Certification Standard.

Describe any critical risks, relating to project implementation, that the stated project's objectives may not be achieved. Detail any risk mitigation measures. Please provide a table with critical risks identified and mitigating actions (template provided).

Critical risks for implementation

Use as many lines as needed.

Description of risk (indicate level of likelihood: Low/Medium/high)	WP(s) involved	Proposed risk-mitigation measures
Scientific: Barren Plateaus persist despite QFF/Evolutionary methods. (Medium)	WP3, WP4	**Mitigation:** If the Forward-Forward approach proves unstable, we will fall back to a "Hybrid Memetic" strategy using standard VQE but optimized via Prof. Acampora's proven Evolutionary Algorithms, which do not rely on gradients.
Technical: Ensemble Aggregation fails to achieve accuracy targets due to weak individual learners. (Medium)	WP2, WP4	**Mitigation:** We will implement advanced "Weighted Voting" mechanisms where the weight of each chip's output is determined by its **QUARK reliability score**. Additionally, we will use the **Quantum Transformer** ansatz, which is designed to maximize expressibility even on small chips.
Technical: Fuzzy-Diffusion fails to capture temporal correlations in EEG. (Low)	WP3, WP4	**Mitigation:** We will pivot from pure Diffusion Models to **Quantum GANs** enriched with Fuzzy Logic Controllers. GANs are notoriously unstable, but the Fuzzy Controller is designed to stabilize learning dynamics, providing a viable alternative generative architecture.
Operational: Delays in accessing sensitive data (LHC/Cyber logs). (Low)	WP4, WP5	**Mitigation:** All partners have direct access (Yoo is CMS member; Lorenz is Fraunhofer Head). However, if delays occur, we have identified open-source proxy datasets (e.g., CERN Open Data Portal, KDD Cup 99) to proceed with algorithmic benchmarking without delay.
Management: IPR disputes regarding industrial exploitation. (Low)	WP1, WP6	**Mitigation:** A comprehensive Consortium Agreement (CA) based on the DESCA model will be signed before the project start, clearly defining

		background/foreground IP and licensing terms for the open-source library.

3.4 Consortium as a whole

(max. 1 page)

The individual members are described in section 3.5, there is no need to repeat that information there.
Describe the consortium. How will it match the project's objectives and bring together the necessary expertise? How do the members complement one another?
In what way does each of them contribute to the project? Show that each has a valid role and adequate resources in the project to fulfil that role.
If applicable, describe the industrial/commercial involvement in the project and explain why this is consistent with and will help to achieve the specific measures which are proposed for exploitation of the results of the project.

Matching Objectives and Expertise

The consortium is designed as a trans-continental feedback loop that bridges the gap between Foundational Quantum Theory (Goals 1.1–1.3) and Rigorous Domain Validation (Goals 2.1–2.3). No single partner possesses the full spectrum of skills required to deliver the "Physics-Aware QML Stack," necessitating this specific collaboration:

- **The "Architects" (SNU):** To achieve **Objective 1 (Scalability)**, we require deep expertise in ansatz design. SNU provides the **Multi-Chip Ensemble** protocols and **Quantum Transformer** architectures needed to distribute learning across fragmented hardware.
- **The "Optimizers" (Univ. Naples):** To achieve **Objective 2 (Trainability)** and **Objective 3 (Robustness)**, we need to solve the Barren Plateau and Noise problems. Univ. Naples injects **Fuzzy Logic** and **Evolutionary/Memetic Algorithms**—classical AI methodologies absent in standard physics labs—to stabilize the quantum models.
- **The "Validators" (Yonsei):** To achieve **Objective 4 (Domain Advantage)**, we need massive, high-dimensional data. Yonsei, through the CMS Collaboration, provides the **LHC particle jet data** that acts as the ultimate stress test for SNU's scalability protocols.
- **The "Certifiers" (Fraunhofer IKS):** To ensure these breakthroughs translate to reality, Fraunhofer IKS provides the **QUARK Framework**. They act as the consortium's "Quality Assurance" department, using QUARK to benchmark the reliability and robustness of all developed algorithms, ensuring they meet industrial standards.

Complementarity and Contribution

The partners complement each other through a "Scale-Logic-Trust" triangle:

- **SNU (Scale)** provides the *structure* of the quantum models. Without SNU, the consortium would lack the scalable architectures (Multi-Chip) to run on NISQ devices.
- **Naples (Logic)** provides the *training engine*. Without Naples, SNU's deep architectures would be untrainable (Barren Plateaus) or fragile to noise. Their **Fuzzy Logic Controllers** provide the "soft" resilience layer.
- **Fraunhofer IKS (Trust)** provides the *standard*. Without Fraunhofer, the results would remain academic proofs-of-concept. The **QUARK framework** transforms the consortium's outputs into certified, reliable software assets ready for safety-critical deployment.
- **Yonsei (Physical Reality)** grounds the project. Their experimental HEP expertise prevents the project from becoming purely theoretical, forcing the algorithms to handle the noise and dimensionality of real subatomic physics.

Industrial/Commercial Involvement

While the core consortium focuses on foundational QPR research, Fraunhofer IKS serves as the bridge to industry. As a leading German RTO (Research and Technology Organisation), Fraunhofer's involvement ensures the path to exploitation is built-in from day one.

- **Consistency with Exploitation Measures:** The proposed **"Quantum Software Reliability Standard"** (WP5) is directly consistent with Fraunhofer's mandate within the **Munich Quantum Valley**. By using the **QUARK framework** to certify the consortium's algorithms, Fraunhofer validates the commercial viability of "Trustworthy QML" services. This allows the consortium to offer not just algorithms, but "Certified Algorithms" to industrial stakeholders (e.g., automotive, finance) who demand reliability guarantees, maximizing the economic impact of the project.

3.5 Description of the consortium

(max. 1 page per partner)

Describe expertise and role in the project for each partner (templates provided). The information provided here will be used to judge the operational capacity. Use the following templates for the coordinator, the other partners requesting funding, and partners not requesting funding if any. If the project relies on input to be provided by a third party, append a letter of commitment at the end of the proposal.

<table>
<tr><td>Jiook Cha
Project Coordinator</td><td>Seoul National University / Department of Psychology, Department of Brain and Cognitive Sciences, Interdisciplinary Program in Artificial Intelligence</td></tr>
<tr><td colspan="2">

Expertise:

Seoul National University (SNU) is South Korea's premier research university. Prof. Jiook Cha's lab specializes in **Quantum Machine Learning (QML) architectures** and their application to **Computational Neuroscience**. The group possesses unique expertise in:

1. **Multi-Chip Ensembles:** Developing scalable approaches to train QML models across multiple chips, applied to image data, EEG/fMRI, and Reinforcement Learning.
2. **Quantum Time-Series Analysis:** Pioneering quantum algorithms for temporal data, including Quantum Transformers for fMRI analysis.
3. **QML for Neuroscience:** Bridging quantum computing with brain science through specialized neural network architectures.

Principal Investigator: Prof. Jiook Cha (Male)

Prof. Cha is a leading expert in applied QML architectures. His research focuses on creating scalable quantum algorithms for high-dimensional biological data.

Relevant Achievements:

- Developing **multi-chip ensembles** approaches and applying them to image data, EEG data, and reinforcement learning environments.
- Developing **quantum time-series transformers** and applying them to fMRI analysis.
- Developing **hybrid quantum temporal convolutional neural network (HQTCN)** algorithms and applying them to EEG analysis.

</td></tr>
<tr><td colspan="2">

Role in project: Coordinator & QML Architecture Lead.

SNU leads WP1 (Management) and drives WP2 (Scalable Architectures) by implementing the Multi-Chip Ensemble protocols. Prof. Cha also leads the theoretical development of the Neuro-Generative validation in WP4, utilizing his expertise in Quantum Transformers and HQTCNs to process EEG/fMRI data.

</td></tr>
</table>

Use this template if this partner is requesting funding

Hwidong Yoo	**Yonsei University / Department of Physics**

Expertise:
Yonsei University is a leading private research university in Korea with a strong legacy in High Energy Physics (HEP). Prof. Hwidong Yoo is a key member of the **CMS Collaboration at CERN**, providing the consortium with direct access to massive-scale "Big Science" data.

1. **High Energy Physics:** Deep expertise in LHC particle jet analysis, tracking, and event classification.
2. **Big Data Analysis:** Infrastructure for processing petabyte-scale datasets from particle collisions.
3. **Quantum for HEP:** Experience in benchmarking quantum algorithms against classical Deep Learning (CNNs/RNNs) for particle physics tasks.

Principal Investigator: Prof. Hwidong Yoo (Male)

Prof. Yoo is an expert in experimental high-energy physics and the application of advanced computing to particle detection. He leads the validation efforts, bridging the gap between abstract quantum algorithms and physical reality.

Relevant Achievements:

- Leadership roles in the CMS Collaboration (CERN) for particle tracking analysis.
- Development of classical Deep Learning benchmarks for LHC jet tagging.
- Pioneering work in applying Quantum SVMs and QNNs to HEP event classification.

Role in project: Domain Validation Lead (HEP).

Yonsei leads **WP4 (High-Complexity Domain Validation)**. They are responsible for curating the LHC datasets, implementing the quantum feature maps for high-dimensional particle data, and rigorously benchmarking the Scalable QML protocols (WP2) against classical SOTA methods.

Jeanette Miriam Lorenz	**Fraunhofer Institute for Cognitive Systems IKS / Department of Quantum Computing**

Expertise:

Fraunhofer IKS is Germany's leading applied research institute for **Safe and Reliable Software Engineering and Artificial Intelligence**. PD Dr. habil. Jeanette Miriam Lorenz's group specializes in **Reliable Quantum Computing for Applications,** focusing on a broad range of topics encapsulating QML**,** quantum optimization, quantum simulations and physics-inspired methods.

1. **Application-driven QML:** Broad expertise in key QML performance aspects, including data-efficiency, impact of feature maps' design, noise stability and generalization.

2. **Quantum Robustness and Safety:** Extensive hands-on experience in QNN certification and verification against noise impacts and cyberattacks, which are critical for the project's cybersecurity validation and "Quality of Service" standards
3. **Benchmarking:** Fraunhofer IKS is one of the European's leaders in application-driven benchmarking of QCs, coordinates the European QC Benchmarking Coordination Committee, and engages in co-developping the QUARK framework.

Principal Investigator: PD Dr. habil. Jeanette Miriam Lorenz (Female)

PD Dr. Lorenz heads the department 'Quantum Computing' at Fraunhofer IKS and is associate professor at LMU Munich. She also leads the consortium focussing on applications of quantum computing in the Munich Quantum Valley. The mission of her research is to enable the use of quantum computing in safety-critical areas such as aerospace, mobility, logistics and medtech.

Relevant Achievements:

- Establishes the guidelines to perform application-centric benchmarking of QC within Europe, e.g. via the EQCBC and the QUARK-framework
- Extensive work on security and safety of QML, e.g. for the BSI (Federal Ministry of Information Security
- Bridging theoretical work with practical aspects of 'getting quantum computing to work'

Role in project: Reliability & Certification Lead

Fraunhofer leads **WP5 (Reliability, Cybersecurity & Certification)** and **WP6 (Dissemination)**. They apply the **QUARK framework** to benchmark the consortium's algorithms, define the **Quantum Reliability Standards**, and conduct the rigorous **Cybersecurity stress-tests** (Adversarial Attacks) to certify the consortium's algorithms.

Giovanni Acampora	**University of Naples Federico II / Department of Physics “Ettore Pancini”**
Expertise:	

The University of Naples Federico II is one of the oldest universities in the world. Prof. Giovanni Acampora's group is globally recognized for **Computational Intelligence** and **Fuzzy Logic**. They bring the critical "classical intelligence" layer to the quantum stack. 1. **Fuzzy Logic Systems:** World-class expertise in Type-2 Fuzzy Sets and Fuzzy Logic Controllers, essential for the project's noise-resilience goals. 2. **Evolutionary Computation:** Deep experience in Memetic Algorithms and Bio-inspired optimization to solve non-convex problems. 3. **Quantum Computational Intelligence:** Pioneering the fusion of Fuzzy Logic with Quantum Computing (Quantum Fuzzy Logic). **Principal Investigator: Prof. Giovanni Acampora (Male)** Prof. Acampora is a Senior Member of IEEE and a pioneer in Quantum Computational Intelligence. His work focuses on hybridizing soft computing with quantum mechanics to solve trainability issues. **Relevant Achievements:** ● Definition of **Quantum Fuzzy Logic** frameworks for uncertainty management. ● Development of **Evolutionary strategies** for optimizing Variational Quantum Circuits. ● Application of **Memetic Algorithms** to prevent Barren Plateaus in QML.
Role in project: Optimization & Intelligence Lead Naples leads **WP3 (Physics-Informed Optimization)**. They provide the **Evolutionary/Memetic algorithms** to drive the QFF training and design the **Fuzzy Logic Controllers** for the Robust Quantum Diffusion models. They also support WP5 in adversarial evolutionary testing.

3.6 Consortium agreement principles (partner's rights and duties, IPR management, data management, open access, GDPR)

(max. ½ page)

Consortium Agreement & Decision Making

The consortium will sign a Consortium Agreement (CA) based on the DESCA (Horizon Europe) model prior to the project start. The CA will define the rights and duties of partners, specifying that SNU (Coordinator) manages the administrative interface, while the General Assembly (one vote per partner) decides on major scientific pivots.

Intellectual Property Rights (IPR) Management

- **Background IP:** Partners retain ownership of their pre-existing know-how. Specifically, **Fraunhofer IKS** retains all rights to the core **QUARK framework**, and **Univ. Naples** retains rights to their pre-existing Fuzzy Logic libraries.
- **Foreground IP:** The new "Physics-Aware QML" algorithms (e.g., Multi-Chip Ensemble protocols, Fuzzy-Diffusion architectures) developed jointly will be jointly owned. We adopt an **"Open Source by Default"** strategy for software outputs, releasing the unified library under an **Apache 2.0 license** to maximize adoption.
- **Commercial Exploitation:** Fraunhofer IKS leads the exploitation strategy. If a patentable invention arises (e.g., a specific industrial control system based on Robust QML), the CA outlines fair access rights and royalty-sharing mechanisms for the inventors.

Data Management & Open Access

- **FAIR Data:** We adhere to the **FAIR principles** (Findable, Accessible, Interoperable, Reusable). All non-sensitive "Challenge Datasets" (LHC jets, synthetic EEG) will be deposited in **Zenodo** with DOIs.
- **Open Access:** The consortium commits to **100% Open Access** for peer-reviewed publications (Green or Gold), ensuring immediate availability on **arXiv**.

GDPR & Ethical Compliance

- **Neuroscience Data:** The processing of EEG/fMRI data (WP4) involves human subjects. **SNU** will ensure all data is **pseudonymized** at the source before sharing within the consortium, in strict compliance with **GDPR** and Korean Personal Information Protection Act (PIPA).
- **Cybersecurity Data:** Network logs used in WP5 will be screened to ensure no personally identifiable information (PII) or critical infrastructure vulnerabilities are exposed.

3.7 Significant facilities and large equipment available to the consortium to perform the project

(max. ½ page)

The consortium brings together world-class infrastructure that enables "Big Science" validation without the need for external hardware procurement.

1. High-Performance Computing & Simulation (SNU & Naples)

- **SNU:** Access to extensive GPU clusters required to simulate the **Multi-Chip Ensemble** protocols. Since we are simulating distributed quantum learning (validating >20 qubits across multiple virtual chips), high-memory classical simulation resources are critical.
- **Univ. Naples:** Dedicated computational clusters optimized for **Evolutionary Algorithms**, which require massive parallel processing capabilities to evolve the populations for the Memetic QFF optimization tasks.

2. "Big Science" Data Infrastructure (Yonsei)

- **Yonsei (CMS Collaboration):** Through Prof. Yoo's leadership in the CMS Collaboration, the consortium has authorized access to the **CERN LHC computing**

grid and petabyte-scale storage systems. This provides the massive **High Energy Physics datasets** (particle jets) required for WP4, which are otherwise inaccessible to standard research groups.

3. Reliability Benchmarking & Quantum Hardware Access (Fraunhofer IKS)

- **The QUARK Framework:** Fraunhofer provides the proprietary **QUARK benchmarking suite**, a significant software facility that automates the assessment of reliability metrics (Lipschitz continuity, robustness). This pre-existing infrastructure accelerates WP5 significantly.
- **Munich Quantum Valley (MQV):** As a key member of MQV, Fraunhofer IKS provides access to a diverse ecosystem of quantum hardware backends (Superconducting, Ion Trap) for validating the **Multi-Chip** protocols on physical NISQ devices where applicable.

3.8 Link with ongoing projects

(max. ½ page)

For each partner indicate (if applicable) the ongoing projects linked to the proposal topic, and their funding sources.

3.9 Financial plan

(max. 1 page)

The resources to be committed for each project partner have to be described in the Electronic Submission System by the coordinator. These resources include: Personnel, Consumables, Equipment, Travel, Subcontracting, Provisions, Licensing fees, other. Justify them here. Both the justification and the information in the system will be communicated to the Evaluation Panel.

4. Ethical issues (max. ½ page)

Describe any foreseeable ethical issue that may arise during the course of the research project. Describe all mitigation strategies employed to reduce ethical risk, and justify the research methodology with respect to ethical issues.

5. References (max. 30 references)

Provide references of articles and publicly available documents directly supporting the proposal.